Online Learning — Lecture Notes, November 12, 2026

Online regression with OGD, Ridge regularization, and inventory decisions

1 Review of the previous session: forecast, then correct

Let us return to our bakery example. At the beginning of day t, we choose a quantity q_t to anticipate the demand y_t, which is still unknown. Once sales have been observed, we measure the error and update our decision for the following day. This is exactly the chronology of *Online Learning*: **decide, observe, incur a loss, then learn**.

To begin with, we can measure the forecasting error using an instantaneous squared loss:

$$\ell_t^{\text{quad}}(q) = \frac{1}{2}(q - y_t)^2.$$

The factor $1/2$ simplifies the gradient calculation without changing the minimizers. We shall first use this loss to learn a regression model, then penalize it with Ridge. Finally, we shall change the loss function to account directly for overstock and shortage costs.

1.1 Online gradient descent with covariates

The idea is to propose a model relating the **covariates** $z_t \in \mathbb{R}^d$, available before sales occur, to the **target** y_t. Consider a linear predictor:

$$\widehat{y}_t = \beta_t^\top z_t, \qquad \beta_t \in \mathcal{K} \subseteq \mathbb{R}^d.$$

The vector β_t contains the model coefficients on day t; $\mathcal{K}$ is the set of admissible parameters, assumed to be closed and convex.

A typical covariate vector is

$$z_t = (1, x_t^1, \dots, x_t^p, y_{t-1}, y_{t-2}, \dots, y_{t-w})^\top.$$

The constant 1 introduces an *intercept*. The variables x_t^j may represent a promotion, the day of the week, or weather information available when the forecast is made. The last w sales provide a fixed-dimensional temporal memory.

We can enrich this vector with a moving average computed from past observations, a seasonal lag y_{t-s} (*seasonal naive*), or Fourier components describing a periodic pattern. There is only one rule to follow: **no feature used to predict y_t may depend on y_t or on a future observation**. Fourier coefficients computed from the very same lags may be redundant with those lags in a linear model; calendar-based seasonal components, however, may provide distinct seasonal information.

The loss is now a function of the *parameter* β:

$$\ell_t^{\text{quad}}(\beta) = \frac{1}{2}(\beta^\top z_t - y_t)^2. \tag{1}$$

Reminder: differentiation in one dimension.

If $f(u) = \frac{1}{2}(au - 1)^2$, with a fixed, the chain rule gives

$$f'(u) = a(au - 1), \qquad f''(u) = a^2 \geq 0.$$

We recover exactly the same mechanism in several dimensions, replacing the scalar coefficient a with the vector z_t.

Proposition 1.1 (Gradient and Hessian of the squared loss)**.** *For the loss in* (1)*,*

$$\nabla_\beta \ell_t^{\text{quad}}(\beta) = (\beta^\top z_t - y_t) z_t, \qquad \nabla_\beta^2 \ell_t^{\text{quad}}(\beta) = z_t z_t^\top \succeq 0.$$

Writing $e_t = \widehat{y}_t - y_t$*, projected OGD takes the form*

$$\beta_{t+1} = \Pi_{\mathcal{K}}(\beta_t - \eta_t e_t z_t), \qquad \eta_t > 0. \tag{2}$$

Proof. Apply the chain rule to $\frac{1}{2}(\beta^\top z_t - y_t)^2$, using $\nabla_\beta(\beta^\top z_t) = z_t$. The Hessian is $z_t z_t^\top$, which is positive semidefinite since $v^\top z_t z_t^\top v = (z_t^\top v)^2 \geq 0$. □

Remark 1.2 (What exactly do we update?). The gradient is taken **with respect to** β, not with respect to the scalar forecast $\widehat{y}_t$. We therefore update β_t. The next day's forecast is then $\widehat{y}_{t+1|t} = \beta_{t+1}^\top z_{t+1}$, provided that z_{t+1} is known before the next day's demand is observed.

1.2 Ridge: penalizing the model coefficients

We can, of course, penalize the loss. The purpose of Ridge is to control the magnitude of the coefficients, especially when the variables are correlated or some directions are poorly observed. This is a **shrinkage of the coefficients**, not of the covariates themselves. Unlike LASSO, Ridge does not generally produce exactly zero coefficients: strictly speaking, it is therefore not a variable-selection method.

Definition 1.3 (Instantaneous Ridge loss)**.** For $\lambda > 0$, consider

$$\widetilde{\ell}_t(\beta) = \frac{1}{2}(\beta^\top z_t - y_t)^2 + \frac{\lambda}{2}\|\beta\|_2^2. \tag{3}$$

Here, the quadratic penalty is added *at every round.*

Since we want to apply OGD, let us calculate the gradient and Hessian:

$$\nabla \widetilde{\ell}_t(\beta) = (\beta^\top z_t - y_t) z_t + \lambda \beta,$$
$$\nabla^2 \widetilde{\ell}_t(\beta) = z_t z_t^\top + \lambda I_d \succeq \lambda I_d.$$

The resulting update is

$$\beta_{t+1} = \Pi_{\mathcal{K}}\left((1 - \eta_t \lambda)\beta_t - \eta_t(\beta_t^\top z_t - y_t) z_t\right). \tag{4}$$

The first term tends to pull the coefficients towards zero; the second corrects the observed forecasting error.

Property 1.4 (Strong convexity and logarithmic regret)**.** *Each loss* $\widetilde{\ell}_t$ *is* λ*-strongly convex. Assume that*

$$\mathcal{K} = \{\beta : \|\beta\|_2 \leq B\}, \qquad \|z_t\|_2 \leq R, \qquad |y_t| \leq Y.$$

Then

$$\|\nabla \widetilde{\ell}_t(\beta)\|_2 \leq G_\lambda := (BR + Y)R + \lambda B.$$

With $\eta_t = 1/(\lambda t)$*, projected OGD satisfies, for every fixed comparator* $u \in \mathcal{K}$*,*

$$\sum_{t=1}^{T} [\widetilde{\ell}_t(\beta_t) - \widetilde{\ell}_t(u)] \leq \frac{G_\lambda^2}{2\lambda}(1 + \log T). \tag{5}$$

In other words, the regret upper bound is of order $O(\log T)$, while the average-regret upper bound is of order $O((\log T)/T)$, faster than $O(T^{-1/2})$.

Caution. This guarantee concerns the **regularized losses** $\widetilde{\ell}_t$, and does not automatically apply to the squared errors alone. To obtain strong convexity in every direction, we penalize all coefficients here, including the intercept. In practice, one may choose not to penalize the intercept, but the theoretical justification must then be adapted.

2 Inventory loss: translating operational costs

This time, we change the **instantaneous loss function**: we therefore change the behavior we want to achieve. The objective is no longer merely to predict a value close to y_t, but to choose a quantity $q_t \in \mathcal{K} = [0, Q_{\max}]$ that accounts for the economic cost of making an error.

Two unit costs are involved: $c_h > 0$ for one excess unit produced (*holding cost*: storage, unsold goods, or waste), and $c_b > 0$ for one unit of unmet demand (*shortage cost*, sometimes called *backorder cost*, depending on the operational setting).

Using $(a)_+ = \max\{0, a\}$, the *newsvendor* loss is

$$\ell_t^{\text{inv}}(q) = c_h(q - y_t)_+ + c_b(y_t - q)_+. \tag{6}$$

We assume here that there is neither inventory carryover nor a replenishment lead time: each day is a single-period decision.

2.1 We recover the pinball loss

Define the **critical ratio**

$$\tau = \frac{c_b}{c_h + c_b} \in (0, 1).$$

Then $1 - \tau = c_h/(c_h + c_b)$, and

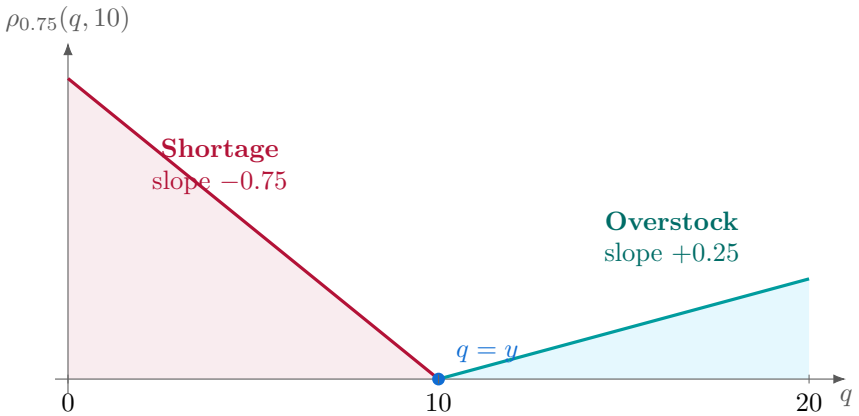
$$\begin{aligned}\ell_t^{\text{inv}}(q) &= (c_h + c_b)\underbrace{\left[(1-\tau)(q - y_t)_+ + \tau(y_t - q)_+\right]}_{\rho_\tau(q, y_t)} \\ &= (c_h + c_b)\rho_\tau(q, y_t).\end{aligned}$$

We can directly recognize the **pinball loss** within the inventory loss. When $\tau = 1/2$, it is proportional to the absolute loss; when $\tau \neq 1/2$, it penalizes overestimation and underestimation differently.

2.2 Visualizing the pinball loss for $\tau = 0.75$

Fix $y = 10$. The normalized loss is piecewise affine:

$$\rho_{0.75}(q, 10) = \begin{cases} 0.75(10 - q), & q < 10, \\ 0, & q = 10, \\ 0.25(q - 10), & q > 10. \end{cases}$$



Underproduction (to the left of y) is penalized three times as heavily as overproduction.

Example 2.1 (A shortage that costs three times as much)**.** If $c_b = 3c_h$, then $\tau = 3/4 = 0.75$. One missing unit costs three times as much as one excess unit. We should therefore order *more* than we would if the two costs were equal: the optimal probabilistic decision targets the 75 % quantile of demand.

This is indeed the direction of the asymmetry: if $\tau > 0.5$, then $c_b > c_h$, and the slope on the left has the greater absolute value. If $\tau < 0.5$, overstock is more costly. When $\tau = 0.5$, the two slopes have the same magnitude: we recover the median case.

3 What is the optimal q? From risk to quantiles

Question: what quantity should we choose *before* demand is observed? In a probabilistic setting, we minimize the **risk**, that is, the expected loss. This differs from minimizing $\ell_t^{\text{inv}}(q)$ after observing y_t, whose minimizer would simply be $q = y_t$: that information is not available when placing the order.

3.1 Reminder: statistical risk and indicator functions

Definition 3.1 (Expected risk)**.** For a predictor $\mu_\theta(X)$ and a loss ℓ, define

$$\mathcal{R}(\mu_\theta) = \mathbb{E}_{(X,Y)}\big[\ell(\mu_\theta(X), Y)\big].$$

An optimal parameter is an element of $\arg\min_\theta \mathcal{R}(\mu_\theta)$, whenever this set is nonempty.

Recall the indicator of an event A:

$$\mathbb{1}_A = \begin{cases} 1, & A \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases} \qquad \text{and} \qquad \mathbb{P}(A) = \mathbb{E}[\mathbb{1}_A].$$

For example, $\mathbb{E}[\mathbb{1}_{\{Y \le q\}}] = \mathbb{P}(Y \le q)$. We shall use this identity to *differentiate* the inventory risk.

3.2 Inventory risk and its derivative

Assume that $Y \ge 0$ is integrable, with cumulative distribution function

$$F_Y(q) = \mathbb{P}(Y \le q), \qquad F_Y(q^-) = \mathbb{P}(Y < q).$$

The inventory risk associated with a constant quantity q is

$$\mathcal{R}(q) = \mathbb{E}[c_h(q - Y)_+ + c_b(Y - q)_+]. \tag{7}$$

> **Caution.** Do not replace the positive parts $(q - Y)_+$ and $(Y - q)_+$ with indicator functions in (7): indicators do not measure the **magnitude** of an overstock or shortage. They appear when we **differentiate** the loss with respect to q.

Indeed, when $q \neq Y$,

$$\frac{\partial}{\partial q}(q - Y)_+ = \mathbb{1}_{\{Y < q\}}, \qquad \frac{\partial}{\partial q}(Y - q)_+ = -\mathbb{1}_{\{Y > q\}}.$$

If the distribution of Y is continuous, we can differentiate under the expectation and obtain

$$\begin{aligned} \mathcal{R}'(q) &= c_h\mathbb{E}[\mathbb{1}_{\{Y<q\}}] - c_b\mathbb{E}[\mathbb{1}_{\{Y>q\}}] \\ &= c_h\mathbb{P}(Y < q) - c_b\mathbb{P}(Y > q) \\ &= c_h F_Y(q) - c_b(1 - F_Y(q)) \\ &= (c_h + c_b)\,(F_Y(q) - \tau)\,. \end{aligned}$$

We obtain a particularly simple expression: the sign of the derivative is determined by the position of $F_Y(q)$ relative to the critical ratio τ.

Proposition 3.2 (The optimal quantity is a quantile)**.** *Let $c_h, c_b > 0$ and $\tau = c_b/(c_h + c_b)$. A quantity $q^\star$ minimizes the risk (7) over $\mathbb{R}$ if and only if*

$$F_Y(q^{\star-}) \leq \tau \leq F_Y(q^\star). \tag{8}$$

One optimal choice is therefore the generalized quantile

$$q^\star = F_Y^{-1}(\tau) := \inf\{q \in \mathbb{R} : F_Y(q) \geq \tau\}.$$

When F_Y is continuous and strictly increasing in a neighborhood of this quantile, we recover the equation

$$F_Y(q^\star) = \tau, \qquad q^\star = F_Y^{-1}(\tau).$$

Proof. The function $\mathcal{R}$ is convex, since it is the expectation of convex functions. Without assuming that Y has a continuous distribution, its one-sided derivatives are

$$\mathcal{R}'_-(q) = (c_h + c_b)F_Y(q^-) - c_b,$$
$$\mathcal{R}'_+(q) = (c_h + c_b)F_Y(q) - c_b.$$

The first-order condition for a convex function is $\mathcal{R}'_-(q^\star) \leq 0 \leq \mathcal{R}'_+(q^\star)$, which is equivalent to (8). □

Remark 3.3 (Capacity constraints and contextual information). If $Y \geq 0$ and $q \in [0, Q_{\max}]$, we can take

$$q^\star_{\mathcal{K}} = \min\{Q_{\max}, F_Y^{-1}(\tau)\}.$$

If covariates X are available, the optimal conditional decision instead targets a quantile of $Y \mid \{X = x\}$: $q^\star(x) = F^{-1}_{Y|\{X=x\}}(\tau)$, under the usual assumptions.

Key takeaway

The *newsvendor* model does not necessarily ask us to predict the **mean** demand. It asks us to choose a **quantile**, whose level $\tau = c_b/(c_h + c_b)$ is determined by operational costs. The more expensive shortages are relative to overstock, the higher the target quantile.

4 Back to Online Learning: learning without knowing the distribution

In an online setting, we do not necessarily know F_Y, nor do we assume that demands are independent or stationary. We choose $q_t \in [0, Q_{\max}]$ before observing y_t, then update the quantity using the revealed loss.

Property 4.1 (Subgradient of the inventory loss). *For* $\ell_t^{\text{inv}}(q) = c_h(q - y_t)_+ + c_b(y_t - q)_+$,

$$\partial \ell_t^{\text{inv}}(q) = \begin{cases} \{-c_b\}, & q < y_t, \\ [-c_b, c_h], & q = y_t, \\ \{c_h\}, & q > y_t. \end{cases}$$

Every subgradient g_t *satisfies* $|g_t| \le G := \max\{c_h, c_b\}$.

Proof. The identity $(a)_+ = (a + |a|)/2$ gives

$$\ell_t^{\text{inv}}(q) = \frac{c_h + c_b}{2}|q - y_t| + \frac{c_h - c_b}{2}(q - y_t).$$

Apply the known subdifferential of the absolute value, then add the derivative of the affine term. □

Initialize $q_1 \in [0, Q_{\max}]$ and, after observing y_t, select $g_t \in \partial \ell_t^{\text{inv}}(q_t)$.

Observed outcome	$\partial \ell_t(q_t)$	g_t	Correction before projection
Shortage: $q_t < y_t$	$\{-c_b\}$	$-c_b$	$q_t + \eta_t c_b$
Exact match: $q_t = y_t$	$[-c_b, c_h]$	0	q_t
Overstock: $q_t > y_t$	$\{c_h\}$	c_h	$q_t - \eta_t c_h$

Definition 4.2 (Projected OGD update).

$$q_{t+1} = \Pi_{[0,Q_{\max}]}(q_t - \eta_t g_t), \qquad \Pi_{[0,Q_{\max}]}(u) = \min\{Q_{\max}, \max\{0, u\}\}.$$

A shortage therefore increases the next order; overstock decreases it. At an exact match, choosing $g_t = 0$ is valid. Note that the correction magnitude *before projection* depends on the step size and the unit cost, **not on the size of the error** $|q_t - y_t|$.

Theorem 4.3 (OGD regret for inventory decisions). *Let* $\mathcal{K} = [0, Q_{\max}]$, *with diameter* $D = Q_{\max}$, *and* $G = \max\{c_h, c_b\}$. *For a constant step size* $\eta > 0$,

$$\text{Reg}_T^{\text{inv}} := \sum_{t=1}^{T} \ell_t^{\text{inv}}(q_t) - \min_{q \in \mathcal{K}} \sum_{t=1}^{T} \ell_t^{\text{inv}}(q) \le \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}.$$

If T *is known and* $\eta = D/(G\sqrt{T})$, *then*

$$\text{Reg}_T^{\text{inv}} \le DG\sqrt{T}, \qquad \limsup_{T \to \infty} \frac{\text{Reg}_T^{\text{inv}}}{T} \le 0.$$

The comparator is the best **fixed quantity**, chosen after observing all T demands: an empirical quantile of order τ, possibly projected onto $[0, Q_{\max}]$. It is not a sequence of conditional quantiles that changes every day. The bound holds without any probabilistic assumption on demand; it is an *upper bound* and does not, by itself, imply two-sided convergence of the realized average regret to zero.

5 Summary for the practical session

We now have three models to compare: **regression OGD**, **regression OGD with Ridge**, and **inventory OGD**. All three follow the same sequential protocol, but they do not optimize the same loss function.

Model	Assumptions and gradient bound	OGD step size	Regret bound
Squared loss	$\|\beta\| \le B$, $\|z_t\| \le R$, $\|y_t\| \le Y$; $G_{\text{quad}} = (BR+Y)R$	$\eta = 2B/(G_{\text{quad}}\sqrt{T})$	$\le 2BG_{\text{quad}}\sqrt{T}$
Ridge	Same bounds, $\lambda > 0$; $G_\lambda = (BR+Y)R + \lambda B$	$\eta_t = 1/(\lambda t)$	$\le \frac{G_\lambda^2}{2\lambda}(1+\log T)$
Inventory	$q \in [0, Q_{\max}]$; $G = \max\{c_h, c_b\}$	$\eta = Q_{\max}/(G\sqrt{T})$	$\le Q_{\max}G\sqrt{T}$

Remark 5.1 (How should we read this table?). The first two rows concern the *coefficients* of a linear predictor; the third concerns the *order quantity* directly. For Ridge, logarithmic regret applies to the **regularized losses**. For both $\sqrt{T}$ bounds, the horizon T is assumed known when choosing the step size. All these bounds are sublinear in T: their upper bounds, divided by T, tend to zero.

Key takeaway

A **loss function** formalizes the objective; its **(sub)gradient** provides the online correction; and **convexity and boundedness assumptions** allow us to control regret. The squared loss targets a mean, the absolute loss a median, and the pinball loss a quantile. In inventory management, the target quantile depends directly on the relative shortage and holding costs.